

UI/UX Specification Document

Simplified Network Monitoring System

Version: 1.0

Date: January 19, 2026

Design System

Color Palette

Primary Colors

```
--primary-50: #FFF6FF;
--primary-100: #DBEAFE;
--primary-500: #3B82F6; /* Main brand color */
--primary-600: #2563EB;
--primary-700: #1D4ED8;
--primary-900: #1E3A8A;
```

Status Colors

```
--success: #10B981; /* Operational/OK */
--warning: #F59E0B; /* Warning state */
--error: #EF4444; /* Error/Critical */
--info: #3B82F6; /* Informational */
```

Neutral Colors (Light Mode)

```
--gray-50: #F9F6F3;
--gray-100: #F3F4F6;
--gray-200: #E5E7EB;
--gray-500: #6B7280;
--gray-700: #374151;
--gray-900: #111827;
```

Dark Mode Colors

```
--dark-bg: #0F172A;
--dark-surface: #1E293B;
--dark-border: #334155;
--dark-text: #F1F5F9;
```

Typography

Font Family:

- Primary: 'Inter', sans-serif
- Monospace: 'JetBrains Mono', monospace

Font Sizes:

```
--text-xs: 0.75rem; /* 12px */
--text-sm: 0.875rem; /* 14px */
--text-base: 1rem; /* 16px */
--text-lg: 1.125rem; /* 18px */
--text-xl: 1.25rem; /* 20px */
--text-2xl: 1.5rem; /* 24px */
--text-3xl: 1.875rem; /* 30px */
```

Spacing System

```
--space-1: 0.25rem; /* 4px */
--space-2: 0.5rem; /* 8px */
--space-3: 0.75rem; /* 12px */
--space-4: 1rem; /* 16px */
--space-6: 1.5rem; /* 24px */
--space-8: 2rem; /* 32px */
```

Border Radius

```
--radius-sm: 4px;
--radius-md: 8px;
--radius-lg: 12px;
--radius-full: 9999px;
```

Shadows

```
--shadow-sm: 0 1px 2px rgba(0,0,0,0.05);
--shadow-md: 0 4px 6px rgba(0,0,0,0.1);
--shadow-lg: 0 10px 15px rgba(0,0,0,0.1);
```

Key Screens

1. Login Screen

Layout:

- Centered card on gradient background
- Logo at top
- Username/password fields
- "Remember me" checkbox
- Login button
- "Forgot password" link

Features:

- 2FA support (show TOTP input after initial login)
- Loading state on button
- Error messages inline
- Password visibility toggle

Visual Elements:

- Animated gradient background

- Glassmorphic card effect
- Smooth transitions

2. Main Dashboard

Layout:

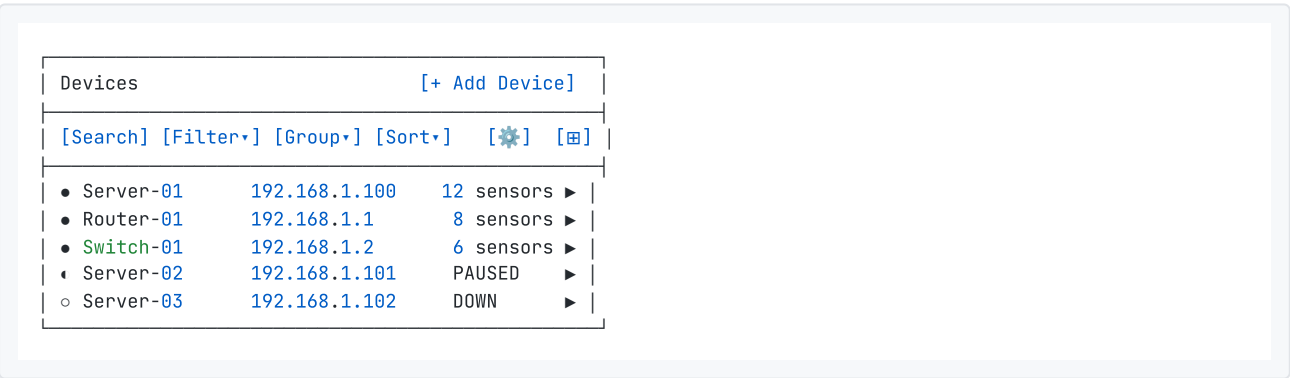


Components:

- **Top Navigation Bar:** Logo, search, notifications, user menu
- **Sidebar:** Collapsible navigation menu
- **Widget Grid:** Drag-and-drop dashboard widgets
- **Quick Stats:** Device count, active alerts, uptime
- **Activity Feed:** Recent events

3. Device List View

Layout:



Features:

- Status indicators (green dot = up, red = down, yellow = warning)
- Real-time status updates
- Bulk selection and actions
- Quick filters (All, Active, Down, Paused)
- Search with highlighting
- Grid/List view toggle

4. Device Detail View

Layout:

← Back to Devices

• Server-01

[Edit] [:]

192.168.1.100 • Ubuntu 22.04 • Datacenter-A

[Overview]

[Sensors]

[Metrics]

[Alerts]

[Logs]

Quick Metrics

CPU: 45%

Memory: 62%

Disk: 38%

Active Sensors

✓ Ping

15ms

Last: 1m

✓ CPU Usage

45%

Last: 30s

⚠ Memory

85%

Last: 30s

Performance Chart (24h)

[Line chart showing metrics over time]

5. Alert Management

Layout:

Alerts

[+ Create Rule]

[Active 5]

[Acknowledged 2]

[Resolved 12]

● CRITICAL

Server-01 CPU > 95%

5m ago

CPU usage at 96%, threshold 95%

[Acknowledge]

[View Device]

⚠ WARNING

Router-01 High Latency

12m ago

Ping response time 250ms, threshold 100ms

[Acknowledge]

[View Device]

Features:

- Color-coded severity badges
- Time since triggered
- Quick acknowledge button
- Alert grouping by device
- Filter by status/severity
- Alert history timeline

6. Dashboard Builder

Layout:

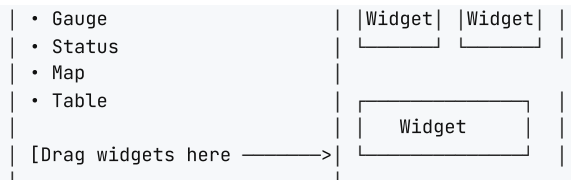
Network Dashboard

[Save] [Cancel]

Widgets:

Canvas

• Chart

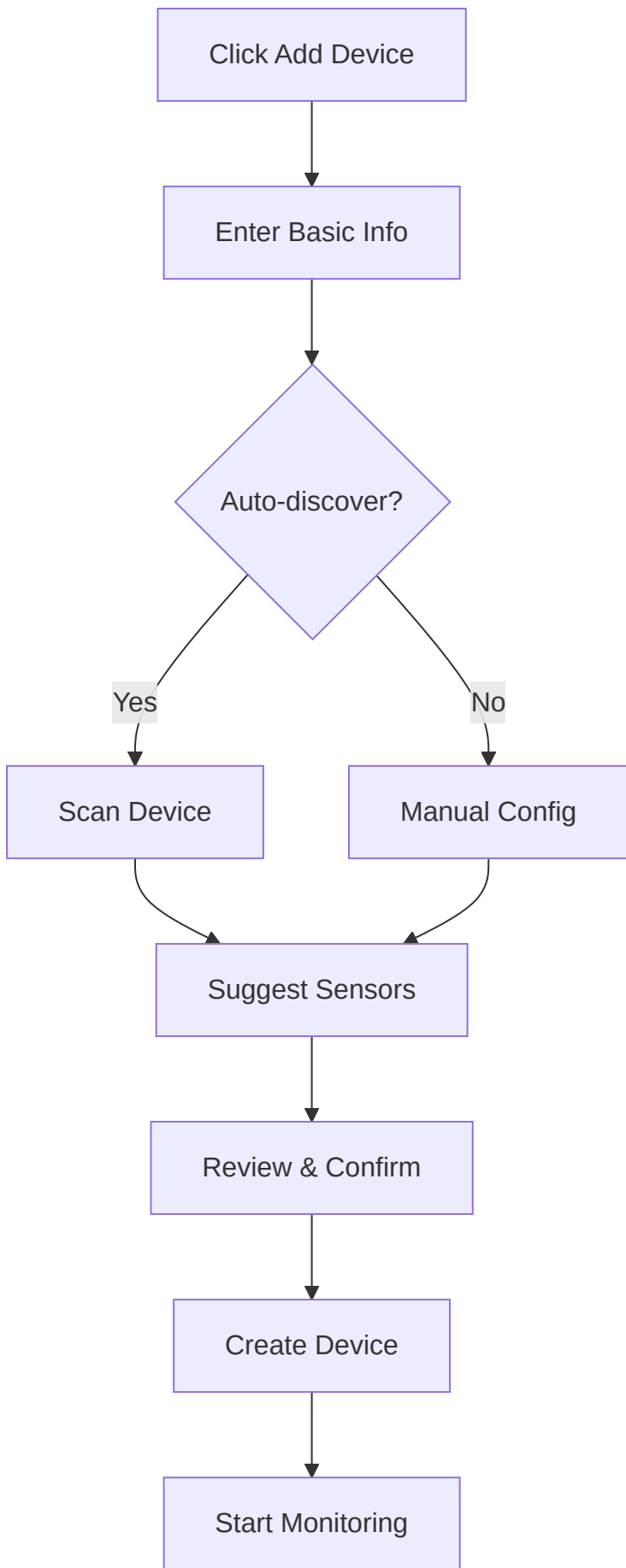


Features:

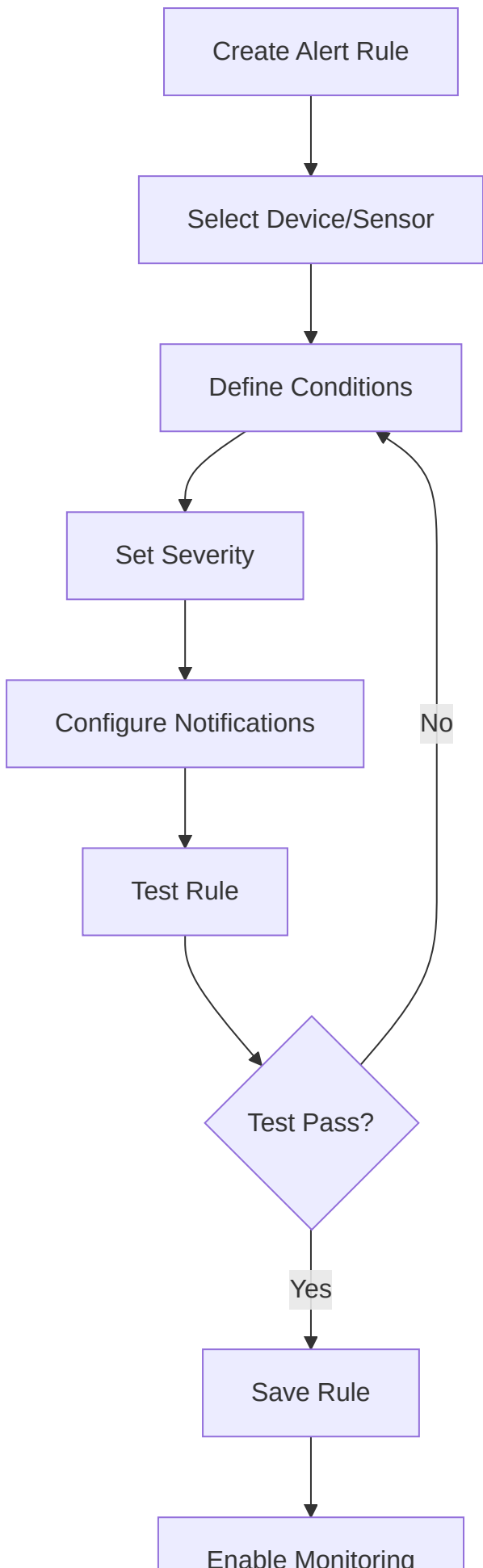
- Drag-and-drop interface
- Live preview
- Widget configuration panel
- Grid snapping
- Responsive layout

User Flows

Device Onboarding Flow



Alert Creation Flow



Components Library

1. Status Badge

● Online

● Warning

● Offline

● Paused

States:

- Online (green)
- Warning (yellow)
- Error (red)
- Offline (gray)
- Paused (blue)

2. Metric Card

CPU Usage

45.2%

Last check: 30s

3. Alert Item

●

High CPU Usage on Server-01

Triggered 5 minutes ago

CPU at 96%, threshold 80%

[Acknowledge] [View Details]

4. Data Table

- Sortable columns
- Row selection
- Inline actions
- Pagination
- Responsive (collapse to cards on mobile)

5. Chart Widget

- Line charts (time-series)
- Bar charts (comparisons)
- Pie charts (distribution)
- Gauge charts (single metric)
- Real-time updates

Responsive Design

Breakpoints

--mobile: 320px - 768px

--tablet: 769px - 1024px

--desktop: 1025px+

Mobile Adaptations

- Collapsible sidebar → Bottom navigation
- Horizontal scrolling for tables → Card layout
- Dashboard grid → Single column
- Reduced widget sizes
- Touch-friendly tap targets (min 44px)

Accessibility

WCAG 2.1 AA Compliance

- **Contrast Ratio:** Minimum 4.5:1 for text
- **Keyboard Navigation:** Full keyboard support
- **Screen Reader:** Semantic HTML, ARIA labels
- **Focus Indicators:** Visible focus states
- **Error Handling:** Clear error messages

Keyboard Shortcuts

- **Ctrl+K**: Global search
- **Ctrl+D**: Add device
- **Ctrl+/'**: Show shortcuts
- **Esc**: Close modals
- **Tab**: Navigate focusable elements

Animations & Transitions

Micro-interactions

```
/* Button hover */
.button {
  transition: all 0.2s ease;
}

/* Card elevation on hover */
.card:hover {
  transform: translateY(-2px);
  box-shadow: var(--shadow-lg);
}

/* Status pulse animation */
@keyframes pulse {
  0%, 100% { opacity: 1; }
  50% { opacity: 0.5; }
}
```

Page Transitions

- Fade in: 200ms
- Slide in: 300ms
- Modal: 250ms ease-out

Dark Mode

Auto-switching

- System preference detection
- Manual toggle in settings
- Per-user preference storage

Color Adjustments

- Inverted backgrounds
 - Reduced chart brightness
 - Adjusted shadows
 - Maintained contrast ratios
-

Notifications

Toast Notifications

✓ Device created successfully

Position: Top-right
Duration: 3-5 seconds
Dismissible: Yes

Types:

- Success (green)
 - Error (red)
 - Warning (yellow)
 - Info (blue)
-

Loading States

Skeleton Screens

Show content placeholders while loading:



Progress Indicators

- Spinner for actions
 - Progress bar for uploads/exports
 - Skeleton for content loading
-

Error States

Empty States



[Add Device]

Error Messages

- Clear description
- Suggested action
- Retry button
- Contact support link

Best Practices

1. **Consistency:** Use design system throughout
2. **Feedback:** Provide immediate visual feedback
3. **Performance:** Lazy load images and charts
4. **Progressive Disclosure:** Show advanced options on demand
5. **Error Prevention:** Validate inputs before submission
6. **Help:** Contextual tooltips and help text